TURKIC INTERNATIONAL MATHEMATICAL OLYMPIAD

TIMO

GRADE 11/12

Mathematical Horizons: Uniting Turkic Minds, Expanding Global Genius.

SCHOLARSHIP FROM TURKISH UNIVERSITIES

MEDIPOL · Bahçeşehir Üniversitesi · İstanbul Üniversitesi · Yıldız Teknik Üniversitesi · İstanbul Galata Üniversitesi

1. In how many different ways can 16 identical balls be distributed in 6 different boxes, provided that each box contains an odd number of balls?

A) 252 B) 244 C) 240 D) 210 E) 180

2. If $x^4 + ax^2 + b$ is divisible by $x^2 + x + 2$ find the value of b–a.

A) –1 B) 0 C) 1 D) 2 E) 3

3. m is an integer which satisfies

$4 \cdot 10^5 < m < 45 \cdot 10^4$

If 20232043 is divisible by m then find the sum of the digits of m.

A)20 B) 21 C) 18 D) 26 E) 24

4. For the polinomial $P(x) = x^3 - 3x^2 - 13x + 15$ to how many of the numbers 2019, 2020, 2021, 2022, 2023, 2024, 2025 and 2026 is the number P(2024) divisible by?

A)2 B) 1 C) 4 D) 5 E) 3

5. Given $0° < x < 90°$ and $\tan x - 2\sin x = 1$
find the value of $4\sin x \cdot \cos x + 1$.

A) $\sqrt{3}$ B) $\sqrt{2}$ C) $\sqrt{5}$ D) $\sqrt{6}$ E) 3

6. How many positive integer pairs (x,y) which satisfy $10x^2 - 6xy + y^2 = 10x - 20$ are there?

A)4 B)2 C) 6 D) 8 E) 0

7. Given the polinomial $P(x) = x^4 + ax^3 + bx^2 + cx + d$ if $P(1) = 10$, $P(2) = 20$, $P(3) = 30$ then find the value of

$$\frac{P(6) + P(-2)}{10} = ?$$

A) 18 B) 52 C) 43 D) 53 E) 13

8. Let A be the set of all functions which satisfy the followings

$f: R \to R,\ f(f(x)) = x^2$ and $f(0) = 0$

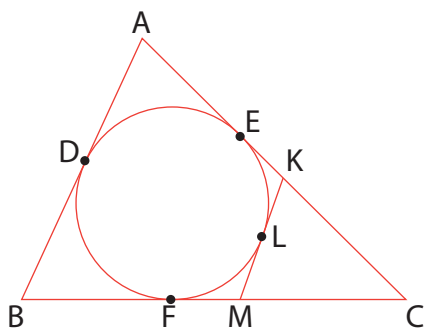
For the function in the set A how many different value can f(1) take?

A) 4 B) 0 C) 1 D) 2 E) 3

9. Given $f(x) = f(x-1) + 3x(x+1)$ and $f(10) = 0$ find the value of $f(1)$.

A) –1315 B) –1314 C) –1316 D) –1318 E) –1312

10.



In the figure above, triangle ABC and its' inner tangent circle are given. The tangent drawn from point L on the arc of the circle to the circle intersects the sides AC and BC at points K and M, respectively. If |AC|=12, |AB|=10 and |BC|=14 then what is the value of the perimeter of the triangle KMC?

A) 8 B) 10 C) 12 D) 14 E) 16

11.

				M				
M	A	T	E	M	A	T	İ	K
	A	T	E	M	A	T	İ	
M	A	T	E	M	A	T	İ	K
				M				

In the table above, the word MATEMATİK will be obtained by moving from one square to the other square from a common corner or a common edge.In how many ways can this be done?

A) 1120 B) 1144 C) 1168 D) 1240 E) 1252

12. a, b and c are relatively prime positive integers and $x = \frac{a}{b}$

If $x + \sqrt{x \cdot (x+3)} + \sqrt{x \cdot (9x+1)} + \sqrt{(x+3)(9x+1)} = 3$

then what is the value of a+b?

A) 17 B) 18 C) 19 D) 20 E) 21

13. Given $0 \le x < 2\pi$, what is the difference between the largest and the smallest of the values of x that satisfy the equation 5cosx + 2sin2x = 4 in radians?

A) $\frac{4\pi}{3}$ B) $\frac{3\pi}{2}$ C) $\frac{5\pi}{4}$ D) $\frac{7\pi}{6}$ E) $\frac{6\pi}{5}$

14. How many different reel numbers x are there that satisfy $\sqrt{x+20} + \sqrt[3]{13 -} = 7 +$ y

A)0 B)1 C) 2 D) 3 E) 4

15. For a convex quadrileteral ABCD if $m(\widehat{ABC})=m(\widehat{ADB})=90^o$, |BC|=|AB|and A(BCD)=36 then what is the value of |BD|?

A) $2\sqrt{3}$ B) $3\sqrt{3}$ C) 12 D) $4\sqrt{3}$ E) 15

16. For all reel numbers $a_1, a_2, a_3, \dots, a_n$ and $b_1, b_2, b_3, \dots, b_n$ the inequality holds.

$(a_1 \cdot b_1 + a_2 \cdot b_2 + \dots + a_n \cdot b_n)^2 \leq (a_1^2 + a_2^2 + \dots + a_n^2)(b_1^2 + b_2^2 + \dots + b_n^2)$

According to this what is the greatest value of the reel number $\sqrt{x-1} + \sqrt{2x-5} + \sqrt{15-3x}$?

A) 5 B) $\sqrt{26}$ C) $3\sqrt{3}$ D) $2\sqrt{7}$ E) $\sqrt{29}$

17. $\lfloor a \rfloor$ is defined as the greatest integer which is not greater than a.
Howmany different reel value of a are there which satisfies the equation $\lfloor a \rfloor \cdot (a - \lfloor a \rfloor) = 2024 \cdot a$?

A)1 B) 2 C) 3 D) 4 E) İnfinitely many

18. How many different positive integer pairs (a,b) which satisfy $a^2+3b^2=1944.a$ are there?

A) 0 B) 1 C) 2 D) 3 E) 4

19. For how many integer values of a the expression $a^8 + 2^{2+2^a}$ is a prime number?

A) 1 B) 2 C) 3 D) 4 E) 5

20 The point O is the centre of the circumcircle of the acute-angled triangle ABC. The K vertex of the square OKLM is on [OC] and the L vertex is on the circle. If the points K, M and A are collinear, what is the measur of the angle $\widehat{ABC}$ in degrees?

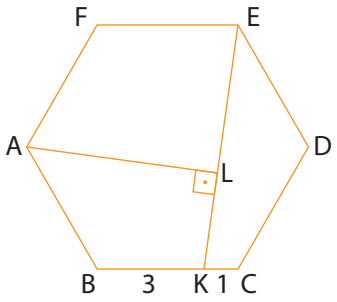
A) 45 B) 52,5 C) 60 D) 62,5 E) 75

21. Given positive reel numbers x, y and z if $x + y + z = 20$
$\sqrt{256 - x^2} + \sqrt{81 - y^2} + \sqrt{16 - z^2} = 21$

Then find the value of $\frac{x+y}{z}$?

A) 6,25 B) 6,5 C) 6,75 D) 7 E) 7,5

22.



In the regular hexagon ABCDEF given above

$[EK] \perp [AL], |KC| = 1$

$|BK| = 3$, $|AL| = \frac{a\sqrt{b}}{c}$ where a, b and c are relatively prime natural numbers.

What is the value of the sum a+b+c?

A) 26 B) 28 C) 29 D) 32 E) 34

23. If the points $A(\sqrt{3}, a)$ *and* $B(\sqrt{3}, b)$ are on the curve $x^4+y^2-1=2x^2y$ then find the value of $|a-b|$.

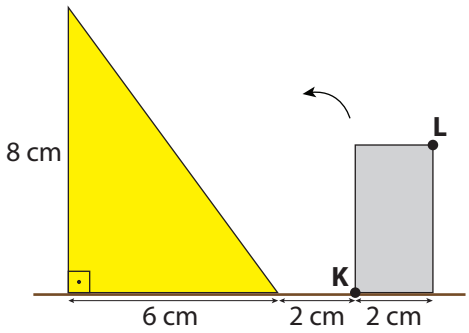
A) 0 B) 1 C) 2 D) 3 E) 4

24. What is the absolute value of the difference in acute angles x that follow the given equation

$\sin^2x + \sec^2x + \tan^2x + \cos^2x + \operatorname{cosec}^2x + \cot^2x = 31$

A) 15° B) 30° C) 45° D) 60° E) 75°

25.



The right side lengths of the yellow right triangle in the figure are 6 cm and 8 cm. The short side of the grey rectangle drawn on the same ground as the triangle is 2 cm. When the rectangle is rotated 90° around point K in the direction of the arrow, point L of the rectangle lies on the hypotenuse of the triangle.
Since the distance between the triangle and the rectangle is 2 cm, what is the area of the region where the triangle and the rectangle intersect after the rectangle is rotated in cm²?

A) $\frac{3}{2}$ B) $\frac{3}{4}$ C) $\frac{5}{2}$ D) 2 E) $\frac{7}{4}$